A THREE-LAYER FRAMEWORK FOR FAKE NEWS DETECTION, VERIFICATION & EXPLANATION

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BACKGROUND & MOTIVATION

REPORT-GROUNDED MOTIVATION

WHY THE PROBLEM MATTERS

False news can spread farther, faster, deeper and more broadly than true news.

WHY CONFIDENCE IS NOT ENOUGH

A high-confidence transformer prediction can still be wrong or poorly calibrated.

PROJECT GOAL

Separate classification, uncertainty and retrieved evidence, then explain those structured signals in plain English.

Motivation is evidence-backed; trustworthiness remains an evaluation goal, not a measured label.

RESEARCH GAP & OBJECTIVES

NO FIELD-WIDE PREVALENCE CLAIMS

RESEARCH GAP

Content-only classification cannot establish whether an article agrees with current external evidence.

A useful system should separate the classifier’s judgement, its uncertainty and evidence retrieved for review.

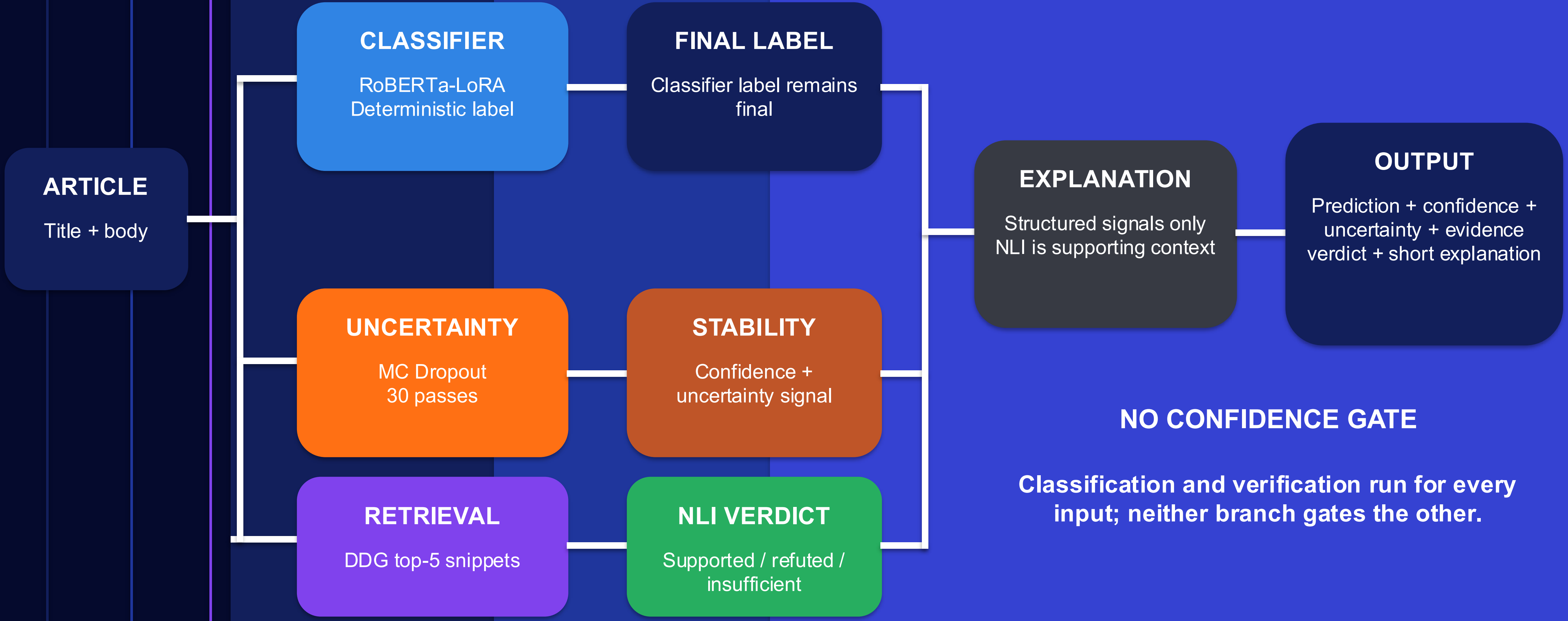
OBJECTIVES

1. Classify full articles with RoBERTa fine-tuned using LoRA.
2. Estimate predictive uncertainty with 30-pass MC Dropout.
3. Retrieve DDG snippets and assign a DeBERTa NLI verdict.
4. Generate a short explanation from structured signals only.

SYSTEM ARCHITECTURE

INDEPENDENT BRANCHES, NO CONFIDENCE GATE

ORIGINAL SLIDE 4



NO CONFIDENCE GATE

Classification and verification run for every input; neither branch gates the other.

DATASET & PRE-PROCESSING

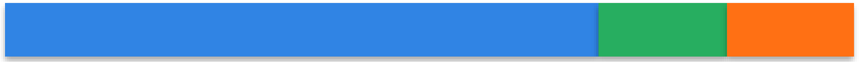
ORIGINAL SLIDE 5

62,649

articles after cleaning

- **72,134** original WELFake rows
- **558** missing titles filled
- **39** rows with missing article text dropped
- **8,618** exact duplicates removed

DETERMINISTIC STRATIFIED SPLIT, RANDOM STATE 42



TRAIN	VALIDATION	TEST
43,854	9,397	9,398

TRAINING SETUP

3	1	2
maximum epochs	early-stop patience	restored epoch

EARLY STOPPING QUANTITY

Validation loss, not training loss, selected the epoch-two adapter.

COMPUTE ENVIRONMENT

Google Colab with one NVIDIA T4 GPU.

MODEL IMPLEMENTATION

CONFIGURATION AND BASELINE PURPOSE

0.2360%

trainable parameters

294,912 trainable

of 124,942,082 total

Frozen RoBERTa base; LoRA updates query/value projections.

KEY CONFIGURATION

Setting	Value
Max length	256
Batch size	32
Learning rate	2e-4
Weight decay	0.01
LoRA dropout	0.05
LoRA rank	8

TF-IDF + LOGISTIC REGRESSION

Purpose

Test whether contextual representations add value beyond weighted word frequencies.

Result boundary

Outperformed TF-IDF on this WELFake split; transfer is untested.

EXPLAINABILITY

ORIGINAL SLIDE 7

EXPLANATION AGENT INPUTS

1. Label
2. Classifier confidence
3. MC uncertainty
4. NLI verdict
5. Max entailment score
6. Max contradiction score
7. Evidence count

INPUT BOUNDARY

Raw article and evidence text are never sent to the explanation agent. Only the seven structured values above are serialised.

REPORT-SUPPORTED BEHAVIOUR

The classifier label is non-negotiable. Conflicting NLI evidence is reported as context, not as grounds for reclassification.

OUTPUT: a short plain-English explanation grounded in the structured signals.

RESULTS AND KEY FINDINGS

HELD-OUT WELFAKE EVALUATION

ORIGINAL SLIDE 8

99.57%
Accuracy

99.57%
Macro F1

40
Errors / 9,398

ERROR PROFILE: TRUE LABELS IN ROWS

Actual \ Predicted	Real	Fake
Real	5,178	15
Fake	25	4,180

BASELINE COMPARISON

Model	Accuracy	Macro F1
RoBERTa-LoRA	**0.9957**	**0.9957**
TF-IDF + Logistic	0.9526	0.9521
Majority class	0.5526	0.3559

KEY FINDING

+4.31 percentage points accuracy and +4.36 points macro F1 versus TF-IDF.

EVIDENCE BOUNDARY

One held-out WELFake split. The report does not claim LIAR or cross-dataset generalisation.

UNCERTAINTY AND NLI VERIFICATION

OFFLINE EVALUATION, NOT DEPLOYED OVERRIDE

UNCERTAINTY SEPARATES ERRORS

Mean entropy: 0.0333 correct vs 0.4816 incorrect.

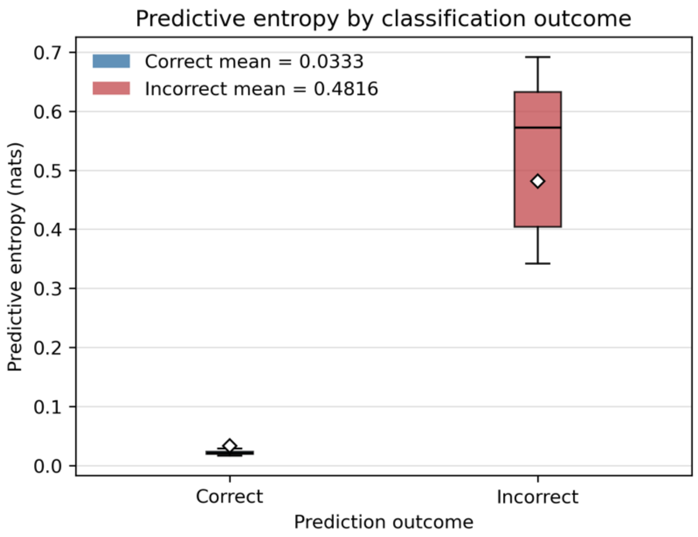
OFFLINE SELECTIVE DEFERRAL

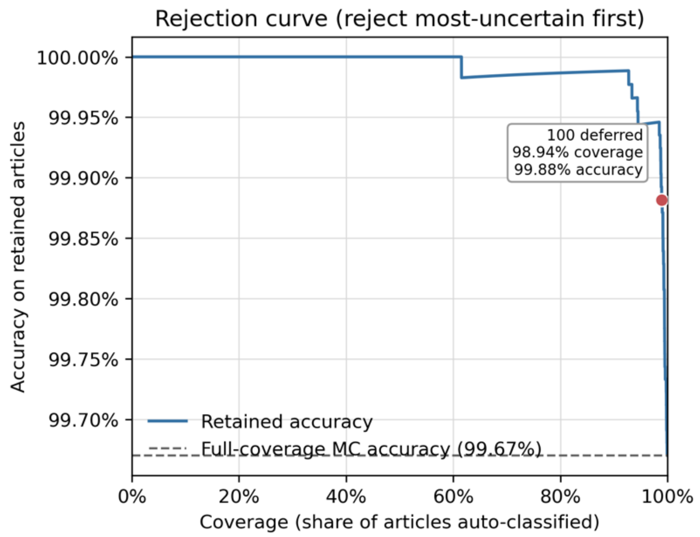
Defer 100 highest-entropy articles: 98.94% coverage, 99.88% retained accuracy.

OFFLINE DIRECT NLI OVERRIDE

100 low-confidence articles: 76% to 64%; 11 fixed, 23 broken.

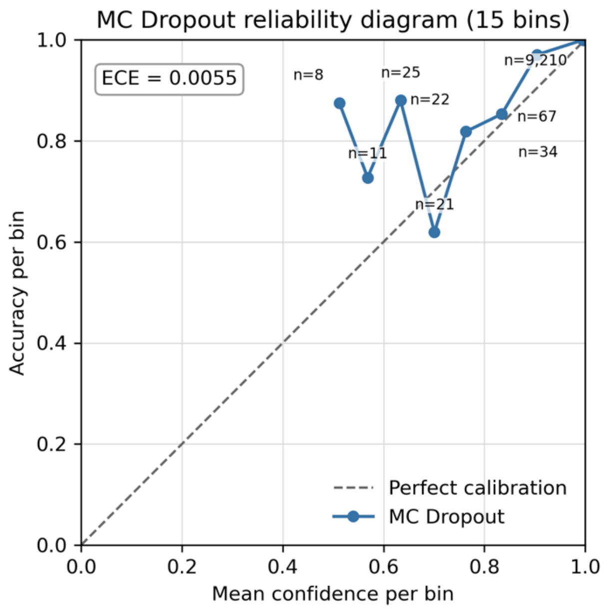
Exact McNemar p = 0.0576; not significant at 0.05





CHALLENGES AND LIMITATIONS

WHAT THE REPORT SUPPORTS AND WHERE IT STOPS



CALIBRATION

MC ECE 0.0055 vs deterministic 0.0028; averaging did not improve calibration.

DDG VERDICT STABILITY

Supported/refuted/insufficient changed 42/32/26 to 30/12/58; both runs were 64/100.

EXPLANATION FAITHFULNESS

8/10 matched signals; 95% Wilson interval 49.0% to 94.3%; one reviewer; two concrete failures.

REPORT-STATED LIMITATIONS

01 One WELFake split; no LIAR or external corpus.

02 Escalation covers 100 of 9,398 low-confidence articles.

03 Live DDG results can change between runs.

04 Faithfulness: n=10, one reviewer, no agreement measure.

CONCLUSION AND FUTURE WORK

Keep the classifier label final; use MC uncertainty for routing and NLI evidence as supporting context for review.

CLASSIFY

0.9957 accuracy and macro F1 on held-out WELFake; no transfer claim.

ROUTE

Entropy supports routing; deferring 100 highest-entropy articles raised retained accuracy.

VERIFY

Direct NLI override reduced observed accuracy from 76% to 64% in the 100-article low-confidence bucket.

NEXT TESTS IDENTIFIED IN THE REPORT

1. **Cross-dataset testing**
2. **Validated routing thresholds**
3. **Stable evidence snapshots**
4. **Larger multi-reviewer explanation study**

PUBLIC GRADIO DEMONSTRATION

REPORT-SUPPORTED DEPLOYMENT CLAIM

DEPLOYED PIPELINE

The public Gradio deployment demonstrates the complete classifier, verification and explanation path.

DEMO VIDEO

https://drive.google.com/file/d/1B2MIXBLEgQGOt1OER63uCFGJny0Qogu6/view?usp=sharing

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COMPLETED AND REPORT-NUMBERED

ORIGINAL SLIDE 14

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